
VITA: ZERO-SHOT VALUE FUNCTIONS VIA TEST-TIME ADAPTATION OF VISION–LANGUAGE MODELS

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ABSTRACT

Vision–Language Models (VLMs) provide a natural substrate for zero-shot goal-conditioned value functions, yet fixed pre-trained representations constrain both distributional generalization and temporal inference. We introduce VITA, a zero-shot value-function learning procedure that augments these capabilities through test-time adaptation. During inference, a lightweight adaptation module is updated by a gradient step on a meta-learned self-supervised objective, with the training objective explicitly optimized so that the post-update representation improves value prediction. Sequential updates along a trajectory store history in the module parameters, thereby providing an implicit memory mechanism for temporal reasoning. To reduce shortcut learning during meta-training, we define a dissimilarity-based sampler that selects semantically diverse trajectory segments. On real-world robotic manipulation tasks, VITA transfers from a single training environment to out-of-distribution tasks, environments, and embodiments, and outperforms the state-of-the-art zero-shot autoregressive-VLM baseline. We further show that VITA value estimates can instantiate dense reward shaping for offline reinforcement learning, yielding multi-task Meta-World policies that outperform policies trained with the simulator’s fuzzy-logic dense rewards.

1 INTRODUCTION

Vision-Language Models (VLMs) exhibit broad task and domain transfer after pre-training on large, weakly structured web corpora without manual annotation (Radford et al., 2021; Alayrac et al., 2022). By comparison, generalist control policies for robotics (Brohan et al., 2023; Kim et al., 2024) and 3D virtual environments (Reed et al., 2022; SIMA Team et al., 2024) remain substantially dependent on expert demonstrations (Ho & Ermon, 2016). World models (Ha & Schmidhuber, 2018; Hafner et al., 2021; Matsuo et al., 2022) address part of this limitation by enabling agents to acquire experience in simulated visual environments, with applications in robotics (Yang et al., 2023; Agarwal et al., 2025), autonomous driving (Hu et al., 2024), and video games (Bruce et al., 2024). Even if such models can synthesize realistic trajectories in diverse, open-ended environments (Hughes et al., 2024; Silver & Sutton, 2025), the central technical question remains: *how can an agent extract scalable learning signals from video without human supervision?*

One research direction learns policies directly from expert visual trajectories by inferring latent actions from visual state transitions (Edwards et al., 2019; Baker et al., 2022; Schmidt & Jiang, 2024; Zhang et al., 2025a). Although these latent actions may subsequently be mapped to executable controls, constructing that mapping remains difficult in practical deployment. A complementary direction learns a universal *goal-conditioned value function* and uses it as a zero-shot reward-shaping signal for reinforcement learning (RL) or as supervision for imitation learning (Chen et al., 2021; Ma et al., 2022). Under this formulation, goal-conditioned value estimation maps a visual observation and a natural-language goal description to a scalar estimate of task progress (Lee et al., 2021a; Ma et al., 2023b; 2024; Dashora et al., 2025).

Prior methods use pre-trained contrastive VLMs (Radford et al., 2021) for zero-shot reward shaping by measuring similarity between task descriptions and visual observations in a shared multimodal embedding space (Ma et al., 2023b; Baumli et al., 2023; Rocamonde et al., 2024). This construction supplies semantic grounding but does not encode the temporal context required to distinguish visually similar states that occur at different task phases, such as folding versus unfolding a shirt.

Autoregressive VLMs (Alayrac et al., 2022) can condition on a full visual trajectory in the prompt, but they may inherit monotonic prediction biases from chronologically ordered pre-training data (Ma et al., 2024). Consequently, both contrastive and autoregressive VLM-based value estimators rely on fixed pre-trained representations for zero-shot prediction (Ma et al., 2023b; 2024; Baumli et al., 2023; Rocamonde et al., 2024; Du et al., 2023), which restricts their generalization and temporal reasoning capacity (Pătrăucean et al., 2023). Existing attempts to improve generalizable value-function learning therefore rely on large-scale pre-training (Alakuijala et al., 2024; Ma et al., 2023b), domain-specific fine-tuning (Zhang et al., 2025b), or demonstration-based reward specification (Sontakke et al., 2023).

We propose VITA, a zero-shot value-function learning method that improves generalization and temporal reasoning through meta-learned test-time adaptation, and that outperforms the state-of-the-art zero-shot approach (Ma et al., 2024) on real-world robotic manipulation tasks. Rather than obtaining transfer through additional large-scale pre-training or task-specific demonstrations, VITA adapts its internal representation online to the semantic and temporal context of each test trajectory. At inference, a lightweight adaptation module receives a gradient update on a meta-learned self-supervised loss (Sun et al., 2024); the meta-objective is chosen so that this update reduces downstream value-prediction error (Finn et al., 2017). Because updates are applied sequentially along the trajectory, trajectory history is accumulated in the module parameters (Sun et al., 2024), yielding an implicit temporal memory. To reduce shortcut learning (Geirhos et al., 2020), we additionally introduce a dissimilarity-based sampling procedure that biases training toward semantically diverse sub-trajectories. We evaluate the resulting estimator along three axes: generalization under task, environment, and embodiment shifts; discrimination between expert and non-expert trajectories; and zero-shot reward shaping for offline RL (Levine et al., 2020).

We summarize our key contributions as follows:

- We propose VITA, a meta-learned test-time adaptation method that equips contrastive VLM representations with zero-shot value estimation, temporal memory, and improved distributional transfer, without task-specific demonstrations or additional large-scale pre-training.
- We show that VITA transfers from a single training environment to out-of-distribution robotic manipulation tasks, environments, and embodiments, while outperforming the state-of-the-art zero-shot baseline (Ma et al., 2024).
- We use VITA value estimates as dense rewards for offline RL and obtain Meta-World MT10 multi-task policies that exceed policies trained with the simulator’s fuzzy-logic dense rewards (Yu et al., 2020).

2 PRELIMINARIES

2.1 GOAL-CONDITIONED VALUE FUNCTIONS

For video trajectories, task-progress estimation can be formalized as goal-conditioned value-function estimation in which reward is identified with task completion. We therefore learn a goal-conditioned value function by predicting the degree of completion from vision–language representations. Formally, $V : \mathcal{O} \times \mathcal{G} \rightarrow [0, 1]$ maps an observation $o_t \in \mathcal{O}$ and a goal specification $g \in \mathcal{G}$ to a scalar progress estimate. We interpret $V(o_t; g) = 0$ as the initial task state and $V(o_t; g) = 1$ as completion. In expert demonstrations, progress is commonly aligned with temporal position under the assumption that expert behavior advances monotonically toward the goal (Lee et al., 2021a; Ma et al., 2024; Dashora et al., 2025). Given an expert trajectory $\tau = (o_1, \dots, o_T) \sim \pi_E$, we define temporal progress by normalized timestep index, $V^{\pi_E}(o_t; g) = \frac{t}{T}$, where T denotes trajectory length. These normalized temporal labels provide supervised targets for learning V .

2.2 TEST-TIME TRAINING FOR SEQUENCE MODELING

Test-time training (TTT) (Sun et al., 2020) treats inference itself as a self-supervised adaptation problem (Goodfellow et al., 2016): model parameters are updated on unlabeled test inputs before producing predictions. Although TTT was introduced for static image classification (Sun et al., 2020), subsequent work extends it to sequence modeling (Sun et al., 2024; Wang et al., 2023) by updating

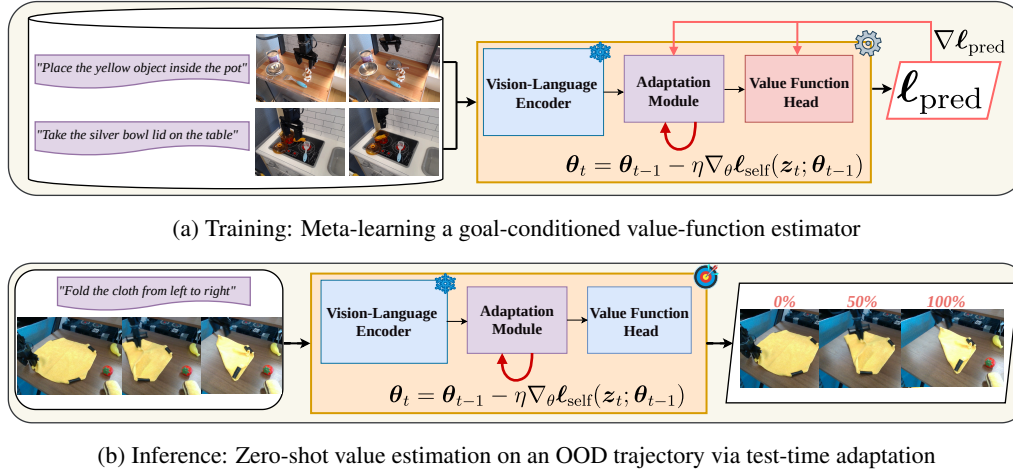


Figure 1: **Overview of VITA.** VITA meta-learns a goal-conditioned value function whose adaptation module is updated at inference, enabling zero-shot value estimation on out-of-distribution trajectories.

a parametric adaptation module f_{adapt} at each timestep with a gradient step on a self-supervised loss. Unlike conventional sequence models, which store temporal information in activation states such as recurrent hidden states (Hochreiter & Schmidhuber, 1997; Cho et al., 2014) or transformer key-value caches (Vaswani et al., 2017), TTT stores history in the parameters of f_{adapt} through sequential inference-time updates. The parameters of f_{adapt} therefore function as implicit memory that accumulates temporal context while preserving chronological order. Sun et al. (2024) further meta-learn the self-supervised task under a gradient-based meta-learning objective (Finn et al., 2017). In that formulation (Sun et al., 2024), learnable linear projections parameterize the self-supervised loss, and the projections are optimized so that a test-time update improves downstream prediction rather than merely minimizing a fixed auxiliary objective. We instantiate this paradigm for goal-conditioned value-function learning, using sequential updates to encode temporal context and meta-learning the self-supervised objective to improve value estimation.

3 VITA: ZERO-SHOT VALUE FUNCTIONS VIA TEST-TIME ADAPTATION

3.1 MODEL ARCHITECTURE

The goal-conditioned value-function estimator consists of three components: (1) a multimodal encoder that maps visual observations and task descriptions into joint representations; (2) an adaptation module whose parameters are updated at test time using a meta-learned self-supervised loss; and (3) a regression head that maps adapted representations to scalar value estimates.

3.1.1 MULTIMODAL INPUT REPRESENTATION

We use the frozen contrastive vision-language encoder CLIP (Radford et al., 2021) to represent both visual observations and language goals. For a visual trajectory $\tau = (o_1, \dots, o_T)$ and task description g , we concatenate the image and text embeddings at each timestep t , obtaining a multimodal sequence $(z_1, z_2, \dots, z_T)$, where $z_t = [\phi_v(o_t); \phi_g(g)] \in \mathbb{R}^{2d}$, and ϕ_v and ϕ_g denote the visual and language encoders, respectively. Because CLIP is trained with a contrastive image-text alignment objective (Radford et al., 2021), observations that are semantically closer to the goal description tend to occupy nearby regions of the shared representation space.

3.1.2 TEST-TIME ADAPTATION

To condition multimodal representations on both semantic content and temporal context, we introduce an adaptation module f_{adapt} following the TTT formulation in Section 2.2. At inference time, the parameters of f_{adapt} are updated online by a gradient step on a meta-learned self-supervised loss ℓ_{self} . This loss is a reconstruction objective defined through learnable linear projections P_K and

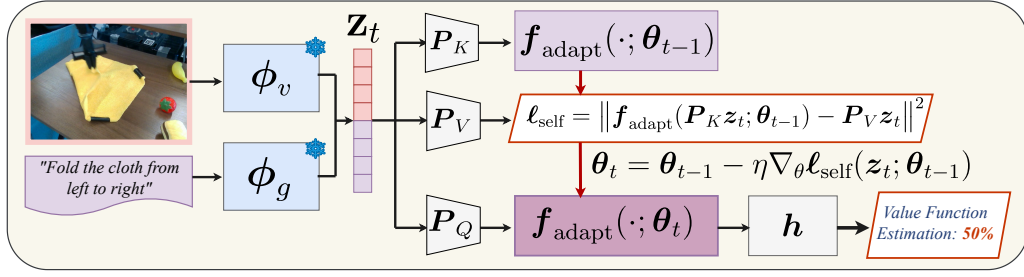


Figure 2: **Test-time Adaptation.** During inference, the adaptation module f_{adapt} receives a gradient update at each timestep t using the meta-learned self-supervised loss ℓ_{self} , thereby encoding temporal history in its parameters.

P_V , which are optimized during training so that the resulting inference-time update improves value estimation. Specifically, $P_K \in \mathbb{R}^{d' \times 2d}$ produces the perturbed input view and $P_V \in \mathbb{R}^{d' \times 2d}$ produces the reconstruction target, yielding

$$\ell_{\text{self}}(z_t; \theta_{t-1}, P_K, P_V) = \|f_{\text{adapt}}(P_K z_t; \theta_{t-1}) - P_V z_t\|^2. \quad (1)$$

where θ_{t-1} denotes the parameters of f_{adapt} before the timestep- t update. The online update rule is

$$\theta_t = \theta_{t-1} - \eta \nabla_{\theta} \ell_{\text{self}}(z_t; \theta_{t-1}), \quad (2)$$

with learning rate η . Because this update is applied sequentially, f_{adapt} accumulates information from previous observations in its parameters and thereby represents temporal context. Section 4.6.2 empirically compares this incremental parameter-memory mechanism with trajectory-level updates.

3.1.3 ZERO-SHOT VALUE FUNCTION ESTIMATOR

After the timestep- t adaptation update, a meta-learned linear projection $P_Q \in \mathbb{R}^{d' \times 2d}$ maps z_t into the adaptation space $\mathbb{R}^{d'}$. The projected representation is processed by f_{adapt} , and a regression head h outputs the predicted goal-conditioned value:

$$V(z_t; g) = h(f_{\text{adapt}}(P_Q z_t; \theta_t)) \quad (3)$$

The regression head h is a two-layer multilayer perceptron (MLP), consistent with value-estimation architectures used in deep reinforcement learning (Espeholt et al., 2018). Training uses the normalized progress labels y_t from expert demonstrations defined in Section 2.1. The supervised prediction loss ℓ_{pred} is the mean squared error between $V(z_t; g)$ and y_t . During inference, h remains fixed and only f_{adapt} is updated by the self-supervised objective, enabling zero-shot value estimation without task-specific test demonstrations.

3.2 TRAINING PROCEDURE

We train the value-function estimator with gradient-based meta-learning. The training objective optimizes the self-supervised loss ℓ_{self} so that the corresponding test-time update decreases the supervised prediction loss ℓ_{pred} (Sun et al., 2024). Mini-batches are constructed from diverse sub-trajectories selected by dissimilarity-based sampling, which reduces reliance on temporal shortcuts and promotes semantic and temporal feature use.

3.2.1 META-LEARNING

During training, f_{adapt} is updated at each timestep using ℓ_{self} . Following gradient-based meta-learning (Finn et al., 2017), we differentiate through these adaptation steps with respect to ℓ_{pred} . This procedure jointly optimizes the initial adaptation parameters θ_0 , the projections P_K, P_V, P_Q , and the regression head h . The total training objective combines ℓ_{pred} and ℓ_{self} , with the latter weighted by λ . The learned θ_0 is then used as the initialization for all inference-time adaptation, so the estimator learns not only a value predictor but also an update rule that improves the predictor on unlabeled test trajectories.

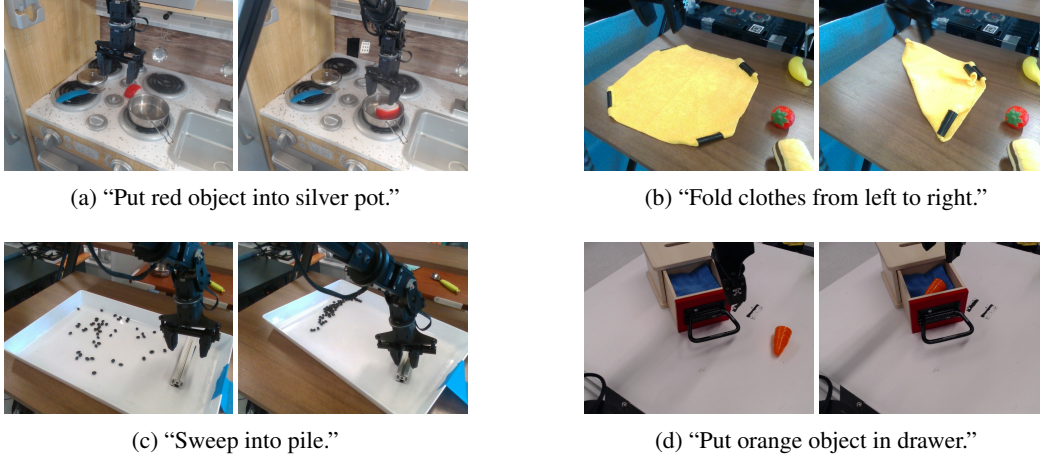


Figure 3: Representative visual trajectories with paired task descriptions under distinct evaluation regimes. (a) In-distribution. (b, c) Environment shift. (d) Joint embodiment and environment shift.

3.2.2 DISSIMILARITY-BASED SAMPLING

Expert visual trajectories typically contain highly redundant adjacent frames, a phenomenon also observed in video action recognition (Wang et al., 2018). This redundancy can induce shortcut learning (Geirhos et al., 2020; Lee et al., 2021b), for example by allowing the estimator to associate frequently observed late-stage visual patterns with high value. We address this issue with a dissimilarity-based sampling procedure that builds mini-batches from visually diverse sub-trajectories within each demonstration. The resulting batches increase intra-batch variation and function as an importance-sampling mechanism that favors semantic diversity.

Given multimodal representations $\{z_1, \dots, z_T\}$, we first extract fixed-length sub-trajectories with a sliding window of size w_{tr} and stride s , producing a candidate set $\mathcal{W}$. The ideal diverse subset of size k solves the pairwise-dissimilarity maximization problem

$$\mathcal{W}' = \arg \max_{\mathcal{U} \subset \mathcal{W}, |\mathcal{U}|=k} \sum_{\{w^i, w^j\} \in \binom{\mathcal{U}}{2}} \|w^i - w^j\|_2^2, \quad (4)$$

where each w^i denotes a length- w_{tr} sub-trajectory. Exact solution of Eq. 4 is combinatorial because it requires enumerating $\binom{|\mathcal{W}|}{k}$ candidate subsets. We therefore use a scoring heuristic in which each window $w \in \mathcal{W}$ receives a score equal to its aggregate dissimilarity from all candidate windows:

$$s(w) = \sum_{v \in \mathcal{W}} \|w - v\|_2^2, \quad \mathcal{W}' = \arg \max_{\mathcal{U} \subset \mathcal{W}, |\mathcal{U}|=k} \sum_{w \in \mathcal{U}} s(w). \quad (5)$$

Selecting the k windows with largest score s yields a polynomial-time approximation to the diverse-subset objective. Appendix G shows that the additional computation is negligible relative to model training. The ablation in Section 4.6.1 compares this sampler with full-trajectory sampling and demonstrates improved expert–non-expert trajectory discrimination.

4 EXPERIMENTS

We evaluate VITA along three functional requirements for goal-conditioned value functions: (i) value-estimation generalization for real-world robotic manipulation under task, environment, and embodiment distribution shifts; (ii) discrimination between expert and non-expert visual trajectories in real-world manipulation settings; (iii) zero-shot dense reward construction for offline RL on Meta-World MT10, a simulated multi-task robot-learning benchmark.

4.1 TRAINING SETUP

We train VITA on expert visual trajectories paired with natural-language task descriptions from BridgeData V2 (Walke et al., 2023), a dataset spanning multiple manipulation tasks, environments, and robot embodiments. The training subset contains 2,986 expert pick-and-place demonstrations and excludes folding, sweeping, and stacking tasks. All training demonstrations use the WidowX 250 embodiment and are collected across 4 ToyKitchen configurations. We reserve 287 additional expert demonstrations as the in-distribution test split, denoted `tk_pnp`. Appendix A provides representative training examples. OpenCLIP ViT-B/32 is used as the frozen encoder for video frames and task descriptions. The training objective combines ℓ_{pred} and ℓ_{self} with self-supervised weight $\lambda_{\text{self}} = 0.5$. For dissimilarity-based sampling, we set the window size to $w_{\text{tr}} = 8$, the number of selected sub-trajectories to $k = 8$, and the stride to $s = 1$. At test time, adaptation uses a single gradient step ($t_{\text{ep}} = 1$) with learning rate $\eta = 0.1$. Appendix D reports additional implementation details and hyperparameter sweeps.

4.2 BASELINES

We compare VITA with value estimators instantiated from contrastive and autoregressive VLMs. VLM-CL computes zero-shot values as cosine similarities between CLIP frame embeddings and task-description embeddings (Baumli et al., 2023). VLM-RM regularizes CLIP embeddings by projecting features along the direction from a generic reference prompt to the task prompt (Rocamonde et al., 2024). CLIP-FT fits a supervised regression head on frozen CLIP multimodal features, whereas CLIP-GRU adds a Gated Recurrent Unit (GRU) (Cho et al., 2014) over the same frozen representations. GVL (Ma et al., 2024) is the autoregressive-VLM zero-shot value-function baseline. We evaluate GVL-0S, in which the prompt contains only the test trajectory and task description, and GVL-1S, in which the prompt additionally contains a shuffled labeled trajectory from the training distribution. Following GVL, we use Gemini 1.5 Pro (`gemini-1.5 Pro-latest`) (Gemini Team et al., 2024) and the original prompt template. We also evaluated Qwen-VL 2.5 (Yang et al., 2024), which did not remove temporal bias, and GPT-4o (Hurst et al., 2024), which declined to produce scalar value estimates. Appendix D gives further baseline details.

4.3 EVALUATING GENERALIZATION UNDER DISTRIBUTION SHIFTS

Table 1: VOC scores for goal-conditioned value estimation under distribution shifts. **ID** = In-Distribution, **ES** = Environment Shift, **EM** = Embodiment Shift, **ES & EM** = Both Shifts.

Shift	Dataset	VLM-CL	VLM-RM	CLIP-FT	GVL-0S	GVL-1S	CLIP-GRU	VITA
ID	tk_pnp	0.038	0.029	0.251	0.269	0.252	<u>0.773</u>	0.782
	lm_pnp	0.017	0.033	0.149	0.305	0.272	<u>0.676</u>	0.725
	td_fold	0.031	0.072	0.152	0.326	0.318	<u>0.674</u>	0.709
	ft_fold	0.108	0.099	0.162	0.331	0.387	0.693	<u>0.658</u>
	rd_fold	0.095	0.055	0.126	0.372	0.406	0.726	<u>0.606</u>
ES	ms_sweep	-0.129	-0.226	0.148	0.158	0.150	<u>0.434</u>	0.490
	dt_tk_pnp	0.042	-0.041	0.149	0.258	0.211	0.856	<u>0.820</u>
	dt_tk_stack	0.035	0.046	0.099	0.254	0.277	<u>0.667</u>	0.708
	dt_ft_stack	0.026	0.028	0.049	0.212	0.249	<u>0.674</u>	0.698
	dt_rd_pnp	0.023	0.041	0.211	0.329	0.316	0.747	<u>0.695</u>
ES & EM								

We measure progress-estimator generalization on curated BridgeData V2 subsets that independently vary task, environment, and robot embodiment. Environment shifts modify scene layout or background: `lm_pnp` is a pick-and-place task in front of a laundry machine; `td_fold`, `ft_fold`, and `rd_fold` are cloth-folding tasks on different surfaces; and `ms_sweep` is a long-horizon sweeping task in a confined tray. Embodiment shifts use the DeepThought robot, which differs from the training embodiment (WidowX 250) in morphology and camera viewpoint. `dt_tk_pnp` and `dt_tk_stack` preserve the ToyKitchen environment while changing embodiment, whereas `dt_ft_stack` and `dt_rd_pnp` combine embodiment and environment shifts. Each dataset contains 200 expert trajectories with task descriptions, except `ms_sweep`, which contains 100 trajectories because of dataset-size

constraints. Appendix B details all evaluation datasets. We report *Value Order Correlation* (VOC) (Ma et al., 2024), which quantifies agreement between predicted progress and chronological frame order. Let $p_1, \dots, p_T$ be predicted progress values for a T -frame trajectory and let $r_t = t$ be the timestep rank. VOC is the Spearman rank correlation ρ_s between $\{p_t\}$ and $\{r_t\}$.

Table 1 shows that VITA and CLIP-GRU consistently exceed GVL-0S and GVL-1S, indicating that explicit preservation of temporal order is critical for progress estimation and that pre-trained autoregressive VLM prompting is insufficient in this setting. Among the two temporally structured methods, VITA obtains higher VOC than CLIP-GRU on 6 of 10 evaluation datasets. Both methods outperform CLIP-based baselines without temporal modeling (CLIP-FT, VLM-CL, VLM-RM), isolating temporal-history encoding as an important factor. CLIP-FT is competitive with GVL on the in-distribution task but does not maintain performance under distribution shift, while VLM-CL and VLM-RM remain weak because their estimates are frame-level similarities. For GVL-0S and GVL-1S, in-context examples produce inconsistent effects: they improve some datasets, have negligible effect on others, and can reduce performance. Both GVL variants perform comparatively well on folding tasks (`td_fold`, `ft_fold`, `rd_fold`) but not on stacking (`dt_tk_stack`, `dt_ft_stack`) or pick-and-place (`dt_tk_pnp`, `lm_pnp`) tasks, suggesting task-type bias in the autoregressive VLM baseline. VITA instead yields stable VOC across task families and shift types, supporting stronger zero-shot generalization than in-context prompting.

The long-horizon `ms_sweep` task reduces VOC for all methods except CLIP-FT; nevertheless, VITA attains the highest score on this dataset. Its advantage over CLIP-GRU suggests that parameter-based temporal memory induced by test-time adaptation is more effective than recurrent hidden-state memory for this long-horizon manipulation setting. Under embodiment shift, VITA also remains robust. On `dt_tk_pnp`, where only the robot embodiment and camera perspective differ from training, both VITA and CLIP-GRU exceed their in-distribution VOC, indicating that temporal-context learning can transfer across embodiments. By contrast, GVL-1S performs poorly on `dt_tk_pnp` and `dt_tk_stack`, showing that the one-shot in-context protocol does not provide reliable embodiment-shift generalization in this evaluation.

4.4 DIFFERENTIATING EXPERT FROM NON-EXPERT TRAJECTORIES

Beyond distributional generalization, we test whether the estimator assigns higher ordered-progress structure to expert trajectories than to suboptimal trajectories. We compare predictions on expert trajectories and scripted non-expert trajectories from the same in-distribution BridgeData V2 setting. The scripted trajectories are produced in ToyKitchen by a heavily randomized controller (Walke et al., 2023) under the same embodiment and environment configuration as the expert demonstrations. The tasks include pick-and-place variants over object categories that may overlap with the training set: general objects (`sc_pnp_obj`), rigid items (`sc_pnp_obj_r`), soft toys (`sc_pnp_stoy`), utensils (`sc_pnp_uten`), and a cloth (`sc_pnp_cloth`). Although these trajectories match the expert data in task family, embodiment, and environment, their execution lacks the smooth monotonic progress characteristic of expert demonstrations. Appendix C shows examples. Because the scripted datasets are generated by randomized control, they do not encode calibrated suboptimality levels. However, a useful value estimator should still produce lower VOC on these trajectories than on expert demonstrations from the same setting. We report *BinVOC*, defined as $\mathbf{1}[\text{VOC}_{\text{exp}} > \text{VOC}_{\text{subopt}}]$; the metric equals 1 when expert trajectories have higher VOC than suboptimal trajectories and 0 otherwise.

We average BINVOC over all scripted datasets. Expert trajectory performance (VOC_{exp}) is computed on `tk_pnp`, the in-distribution pick-and-place test set that matches the scripted evaluation task family. VITA, GVL-0S, and GVL-1S achieve perfect discrimination by assigning higher VOC to expert trajectories for every scripted dataset. VITA exceeds CLIP-GRU, suggesting that recurrent hidden-state memory is more susceptible to temporal shortcut overfitting than the implicit parameter memory produced by sequential test-time updates. The frame-level CLIP baselines (VLM-CL, VLM-RM) underperform because they do not model temporal structure. CLIP-FT is more reliable but fails on one scripted task, consistent with sensitivity to object-distribution changes.

Method	BinVOC
VLM-CL	0.40
VLM-RM	0.00
CLIP-FT	0.80
GVL-OS	1.00
GVL-1S	1.00
CLIP-GRU	0.80
VITA	1.00

Method	IQM	95% CI
VLM-CL	0.760	[0.722, 0.791]
VLM-RM	0.746	[0.718, 0.771]
CLIP-FT	0.785	[0.759, 0.809]
CLIP-GRU	0.777	[0.734, 0.814]
META-WL	0.779	[0.750, 0.804]
VITA	0.815	[0.785, 0.838]

(a) Expert–Non-Expert Discrimination (Bin-VOC).

(b) MT10 Offline RL Performance (IQM).

Table 2: (a) Expert-scripted trajectory discrimination, measured as average BinVOC over 5 in-distribution scripted datasets. (b) Meta-World MT10 offline RL performance, reported as IQM with 95% stratified bootstrap confidence intervals.

4.5 ZERO-SHOT REWARD SHAPING FOR OFFLINE RL

We evaluate zero-shot reward shaping on Meta-World MT10 (Yu et al., 2020), which contains ten robotic manipulation tasks including pick-and-place, door opening, and pushing. In this experiment, VITA is trained only on real-world robotic data and then applied without fine-tuning to the simulated Meta-World domain. For each MT10 task, we generate 20 expert visual demonstrations using the Meta-World expert policies. The goal-conditioned value estimator assigns a dense reward to each visual observation, and Implicit Q-Learning (IQL) (Kostrikov et al., 2021) trains offline policies from the resulting rewards. We repeat expert-data generation and policy training for 10 random seeds. After training, each multi-task policy is evaluated on 20 rollouts per task, and the average episodic return is computed for each task–seed pair. Following Agarwal et al. (2021), we report the interquartile mean (IQM) over all pairs with 95% stratified bootstrap confidence intervals. Appendix E specifies the complete training and evaluation protocol.

Table 2 reports MT10 performance for all scalable baselines under this protocol. VITA obtains the largest IQM return (0.815), exceeding all CLIP-based reward estimators. A full-scale comparison with GVL is not computationally practical because GVL depends on Gemini-1.5, a proprietary autoregressive VLM with prohibitive inference cost for RL reward generation; in preliminary experiments, open-source autoregressive alternatives did not provide reliable progress estimates. We also include the Meta-World fuzzy-logic dense reward (META-WL), which achieves 0.779. VITA surpasses this simulator-provided reward, indicating that a value estimator trained on real-world demonstrations can transfer to simulated dense reward construction.

4.6 ABLATION STUDIES

4.6.1 EFFECT OF DISSIMILARITY-BASED SAMPLING ON DISCRIMINATIVE PERFORMANCE

We isolate the effect of dissimilarity-based sampling by varying the training-time sub-trajectory selection rule and measuring expert–non-expert discrimination. Full-trajectory sampling (Sun et al., 2024) evaluates the adaptation loss over every overlapping sub-trajectory with stride $s = 1$; in our setting, this exposes the model to global temporal regularities that can become shortcuts. Random sampling increases coverage by selecting sub-trajectories uniformly but does not explicitly optimize semantic diversity. Dissimilarity-based sampling selects sub-trajectories with high pairwise distance in representation space and therefore directly promotes semantic variation. This selection rule outperforms both full-trajectory and random sampling for expert–non-expert discrimination. Appendix F.1 provides the full ablation.

4.6.2 EFFECT OF TEMPORAL MEMORY ON VALUE FUNCTION ESTIMATION

We next evaluate how the mechanism used to encode temporal context affects value estimation. VITA performs sequential adaptation without resetting the adapter, updating from θ_{t-1} at each timestep; this implicit-memory procedure stores both temporal order and accumulated context in the parameters. We compare against three alternatives: (i) *trajectory-level adaptation (TTT-TR)*, which

computes one gradient update over the full trajectory and therefore averages the loss across frames, eliminating temporal order; (ii) *memoryless adaptation (TTT-RS)*, which resets the adapter to θ_0 at every timestep and adapts only to the current observation, preventing temporal carryover; (iii) *explicit memory (TTT-EX)*, which also resets the adapter at every timestep but computes its update from a local window of recent frames with the meta-training window size $w_{tr} = 8$. Across all datasets, VITA outperforms the trajectory-level, memoryless, and explicit-memory variants, showing that incremental parameter accumulation is the most effective temporal-context mechanism. Appendix F.2 contains the full analysis.

5 RELATED WORK

5.1 VLMS AS VALUE FUNCTION ESTIMATORS

Contrastive VLMS have been used as zero-shot reward models (Baumli et al., 2023; Rocamonde et al., 2024) and goal-conditioned value functions (Ma et al., 2023b; 2024) by converting frame-text similarity into progress estimates. These methods provide semantic grounding but do not model temporal dependencies. Other work fine-tunes CLIP on domain-specific video datasets (Fan et al., 2022; Jiang et al., 2024) to construct dense rewards, but the required supervision limits zero-shot applicability. RoboCLIP (Sontakke et al., 2023) supports one-shot reward specification without fine-tuning, although its rewards are sparse and still depend on demonstrations. Large-scale multimodal pre-training methods (Alakuijala et al., 2024; Li et al., 2024; Ma et al., 2023b) learn language-conditioned value functions by relying on data scale for transfer. ReWiND (Zhang et al., 2025b) trains a cross-modal sequential transformer on augmented large-scale data with video rewinding and uses a small number of in-domain demonstrations to improve generalization. In contrast, VITA adapts a pre-trained contrastive vision-language representation during inference, using meta-learned test-time updates to encode semantic and temporal context without additional large-scale pre-training or domain-specific demonstrations. The method is compatible with any pre-trained vision-language representation that supports universal value estimation and can improve such representations through meta-learned adaptation. Autoregressive VLMS (Alayrac et al., 2022) perform trajectory-level in-context learning and have been applied to success detection (Du et al., 2023) and goal-conditioned value estimation (Ma et al., 2024). GVL (Ma et al., 2024) reduces monotonic bias from ordered pre-training data by shuffling frames at inference, but this operation removes chronological order, which is necessary for temporal reasoning and for distinguishing visually similar states at different task phases. VITA preserves chronological order and instead encodes temporal context through sequential test-time adaptation.

5.2 TEST-TIME ADAPTATION

Test-time adaptation provides parameter-efficient mechanisms for modifying VLM behavior without full model fine-tuning. Prompt-based methods optimize parameters at test time using task context, but their adaptation mechanisms are often difficult to interpret (Zhou et al., 2022; Khattak et al., 2023; Ma et al., 2023a). Lim et al. (2025) argue that CLIP representations are suitable for test-time adaptation because they encode human-interpretable concepts identified by mechanistic interpretability (Bricken et al., 2023). CLIP-derived similarity has also served as a reward signal whose parameters are optimized at test time through reinforcement learning (Zhao et al., 2024). Test-time training (Sun et al., 2020) instead updates an adaptation module at each timestep using a self-supervised objective. Unlike meta-reinforcement learning methods (Finn et al., 2017; Bauer et al., 2023), which require task-specific adaptation episodes during training, TTT supports online inference-time adaptation without task labels.

6 DISCUSSION, LIMITATIONS, AND FUTURE WORK

6.1 DISCUSSION

Our results show that meta-learned test-time adaptation improves goal-conditioned value estimation for robotic manipulation under task, environment, and embodiment shifts. VITA also discriminates expert from non-expert visual trajectories and supplies dense rewards for zero-shot offline RL, indicating utility beyond standalone progress prediction. The sequential update rule stores trajectory

history in adapter parameters, allowing the estimator to combine semantic information from the VLM backbone with temporal context accumulated during inference. This mechanism yields stronger performance than frame-level CLIP baselines and autoregressive-VLM in-context learning. VITA also improves over CLIP-GRU on expert–non-expert discrimination and offline RL, suggesting that implicit parameter memory generalizes more reliably than recurrent hidden-state memory for zero-shot value estimation. Although pre-trained autoregressive VLMs such as GVL have broad representational capacity, they are not optimized for progress estimation, which requires temporally grounded reasoning over visual trajectories. In our implementation, per-timestep adaptation has negligible overhead because only a lightweight module is updated, so the procedure does not materially affect real-time applicability.

6.2 LIMITATIONS AND FUTURE WORK

Although test-time adaptation improves zero-shot progress estimation across unseen tasks and environments in our experiments, the method may be less reliable for trajectories with high execution variability or substantially longer horizons. The computational overhead of the adapter update is small, but modifying value-estimator parameters at every timestep can introduce deployment risk in safety-critical closed-loop systems, and this issue requires additional study before real-time control deployment. Future work should investigate alternative test-time adaptation mechanisms for vision–language models, especially for agents trained inside world models. Applying VITA to closed-loop robotic control and more complex RL environments is another direct extension. Because the method is agnostic to the specific vision–language encoder, it could also improve the generalization and temporal reasoning of other domain-specific multimodal representations through meta-learned adaptation. Finally, our experiments indicate that dissimilarity-based sampling reduces shortcut learning, but a formal analysis of how diversity-based selection changes shortcut reliance remains outside the present scope.

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A TRAINING DATASET

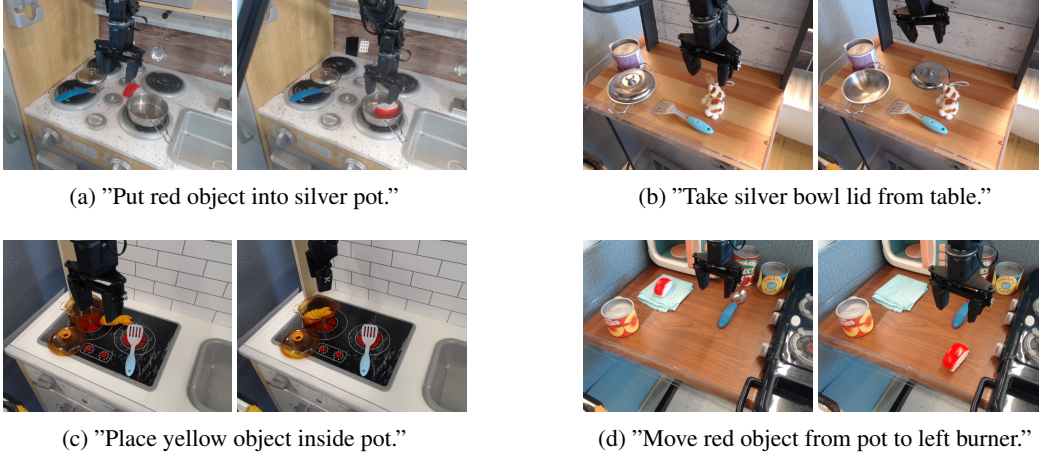


Figure 4: Each subfigure presents the initial and terminal frames of a training expert demonstration together with its natural-language task description. The demonstrations are collected across four ToyKitchen environment configurations.

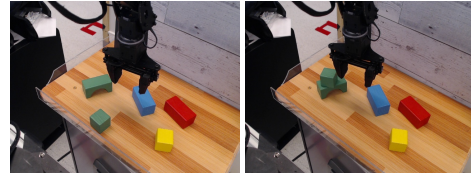
B EVALUATION DATASETS

Table 3: Evaluation dataset specification by task type, environment, and robot embodiment.

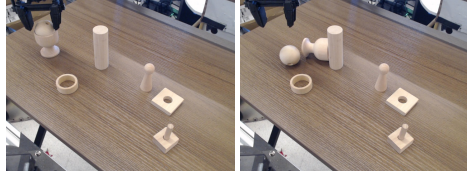
Dataset	Task Type	Environment	Embodiment
tk_pnp	pick-and-place	toy kitchen	WidowX 250
lm_pnp	pick-and-place	laundry machine	WidowX 250
td_fold	fold cloth	tabletop (dark wood)	WidowX 250
ft_fold	fold cloth	folding table	WidowX 250
rd_fold	fold cloth	robot desk	WidowX 250
ms_sweep	sweep	folding table (tray)	WidowX 250
dt_tk_pnp	pick-and-place	toy kitchen	DeepThought
dt_tk_stack	stack blocks	toy kitchen	DeepThought
dt_ft_stack	stack blocks	folding table	DeepThought
dt_rd_pnp	pick-and-place	robot desk (drawer)	DeepThought



(a) dt_tk_ms: "Move yellow object from burner to center."



(b) dt_tk_stack: "Place green cube on top of arch."

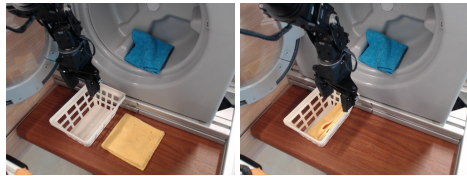


(c) dt_ft_stack: "Move egg to table."



(d) dt_rd_pnp: "Put orange object in drawer."

Figure 5: Each subfigure presents the initial and terminal frames of an evaluation trajectory under embodiment shift with the corresponding natural-language task description. The top row retains the ToyKitchen environment and changes only the robot to DeepThought; the bottom row additionally changes the environment.



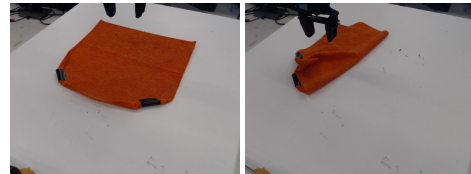
(a) lm_pnp: "Place blue cloth inside washer."



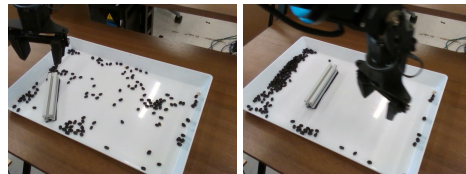
(b) td_fold: "Fold cloth bottom right to top left."



(c) ft_fcnp: "Fold clothes from left to right."



(d) rd_fold: "Fold cloth bottom right to top left."



(e) ms_ft_sweep: "Sweep into pile."

Figure 6: Each subfigure presents the initial and terminal frames of an evaluation trajectory under environment shift with the associated natural-language task description. All trajectories use the same robot embodiment while varying visual appearance and workspace structure.

C SCRIPTED DATASET



(a) `sc_pnp_obj`: "Pick up an object and place it."



(b) `sc_pnp_robj`: "Pick up a rigid object and place it."



(c) `sc_pnp_stoy`: "Pick up a soft object and place it."



(d) `sc_pnp_uten`: "Pick up a kitchen utensil and place it."



(e) `sc_pnp_cloth`: "Move the cloth."

Figure 7: Each subfigure presents the initial and terminal frames of a scripted evaluation trajectory collected in ToyKitchen with the WidowX 250 robot. The tasks instantiate pick-and-place behavior over different object categories. Because public annotations are unavailable, we use generic task descriptions.

D MODEL ARCHITECTURE

D.1 MODEL AND TRAINING DETAILS

We use the OpenCLIP ViT-B/32 encoder pre-trained on the OpenAI dataset as a frozen representation backbone. Visual observations and task descriptions are encoded by the CLIP vision and text encoders, respectively. Based on validation performance, we use a joint concatenation-based representation rather than an element-wise product. The adaptation module f_{adapt} is a two-layer residual MLP with GELU activations and projection dimension $d' = 64$. Optimization uses AdamW with learning rate 1×10^{-4} , weight decay 1×10^{-4} , and a cosine learning-rate schedule with 10% warmup. We train with batch size 32 and pad trajectories to a maximum length of 120 frames, matching the longest training sequence. The self-supervised loss coefficient λ_{self} is set to 0.5 based on validation performance. Training runs for 5 epochs because validation VOC typically saturates early and longer training does not improve performance. Unless otherwise specified, dissimilarity-based sampling uses stride $s = 1$, sub-trajectory length $w_{\text{tr}} = 8$, and $k = 8$ selected windows. All experiments were executed on NVIDIA RTX 6000 Ada Generation GPUs.

D.2 TEST-TIME TRAINING HYPERPARAMETERS

At inference time, only the temporal adaptation module f_{adapt} is updated, with the projection dimension fixed to the training value $d' = 64$. The default setting applies one gradient step ($t_{\text{ep}} = 1$) with learning rate $\eta = 0.1$. We select this configuration from a sweep over learning rates $\{5.0, 1.0, 0.1, 0.05, 0.01, 0.001\}$ and adaptation-step counts $\{1, 5, 10\}$, using the validation

Table 4: Effect of test-time learning rate η and adaptation-step count t_{ep} on validation-set VITA performance measured by VOC.

η	$t_{\text{ep}} = 1$	$t_{\text{ep}} = 5$	$t_{\text{ep}} = 10$
0.001	0.225	0.345	0.453
0.01	0.454	0.744	0.781
0.05	0.746	0.670	0.515
0.1	0.783	0.512	0.369
1.0	0.299	0.183	0.181
5.0	0.169	0.182	0.157

results in Table 4. The sweep indicates that a single larger update ($\eta = 0.1$, $t_{\text{ep}} = 1$) and multiple smaller updates ($\eta = 0.01$, $t_{\text{ep}} = 10$) produce comparable validation VOC. We use the single-update setting because it is computationally cheaper and slightly higher-performing.

D.3 BASELINES

VITA uses $t_{\text{ep}} = 1$, projection dimension $d' = 64$, and learning rate η of 0.1, selected by the hyperparameter sweep described above. CLIP-FT retains the same frozen CLIP encoder and prediction architecture but removes the adaptation module and self-supervised loss. Specifically, it applies a linear projection and two-layer MLP to frozen multimodal features and trains with ordinary supervised regression, without meta-learning or test-time adaptation. To provide additional capacity, CLIP-FT uses an $8\times$ larger projection matrix and $10\times$ more training steps than VITA (Lin et al., 2022). For VLM-RM, the baseline prompt describes the environment. CLIP-GRU also follows the VITA backbone and regression head h , but replaces f_{adapt} with a single-layer GRU that encodes temporal history in recurrent hidden states. The GRU hidden size is 64, and its final hidden state is provided to h . For both GVL-0S and GVL-1S, we use Gemini 1.5 Pro, gemini-1.5-pro-latest (Gemini Team et al., 2024); the original GVL implementation used the earlier gemini-1.5-pro release. We also tested the open-source autoregressive VLM Qwen-VL 2.5 (Yang et al., 2024), which retained temporal bias despite frame shuffling, and GPT-4o (Hurst et al., 2024), which consistently declined to output scalar progress values. All GVL evaluations use the prompt template from the original GVL work (Ma et al., 2024).

E OFFLINE RL SETUP

E.1 TRAINING CONFIGURATION

We use Implicit Q-Learning (IQL) with hyperparameters fixed across all tasks. Expectile regression uses $\tau = 0.7$, advantage weighting temperature is 3.0, and the clipping threshold is 100.0. The actor and critic learning rates are 1×10^{-4} and 3×10^{-4} , respectively, and the batch size is 256. Policies are trained for 100,000 gradient steps and evaluated every 10,000 steps on 20 rollouts per task with horizon 150. The implementation follows d3rlpy with configuration expectile=0.7, weight_temp=3.0, max_weight=100.0, actor_lr=1e-4, critic_lr=3e-4, batch_size=256.

E.2 EVALUATION PROTOCOL

Following Agarwal et al. (Agarwal et al., 2021), we report the interquartile mean (IQM) of returns over all task–seed pairs. IQM averages the middle 50% of scores and is therefore less sensitive to outliers than the arithmetic mean. We estimate 95% confidence intervals with 10,000 stratified bootstrap resamples, stratifying by task. The same aggregation and uncertainty-estimation procedure is applied to every method in the Meta-World MT10 experiments.

F ABLATION STUDIES

F.1 EFFECT OF DISSIMILARITY-BASED SAMPLING ON DISCRIMINATIVE PERFORMANCE

Table 5: Disc@VOC results on scripted datasets for training-time sub-trajectory sampling strategies used in test-time adaptation. A checkmark denotes successful expert-suboptimal trajectory discrimination. Sampling methods **FT** (full-trajectory), **Rand** (random), and **Diss** (dissimilarity-based) are parameterized by (w_{tr}, k) .

Dataset	FT (8,-)	Rand (8,8)	Diss (8,4)	Diss (4,8)	Diss (8,8)
sc_pnp_obj		✓	✓		✓
sc_pnp_robj	✓	✓	✓	✓	✓
sc_pnp_stoy			✓		✓
sc_pnp_uten	✓	✓	✓	✓	✓
sc_pnp_cloth		✓	✓	✓	✓
Avg. Disc@VOC	0.20	0.8	1.0	0.60	1.0

We evaluate dissimilarity-based sampling by measuring how the training-time sub-trajectory selection rule affects expert-suboptimal discrimination. Table 5 reports Disc@VOC on 5 scripted datasets, where success indicates that the model assigns higher average VOC to expert trajectories from `tk_pnp` than to the corresponding scripted evaluation trajectories. We compare three sampling strategies: (1) full-trajectory sampling, which applies adaptation to all overlapping length- w_{tr} sub-trajectories with stride s , following prior work (Sun et al., 2024); (2) random sampling, which uniformly selects k length- w_{tr} sub-trajectories per video; and (3) dissimilarity-based sampling, which constructs the same candidate set and selects k sub-trajectories that maximize pairwise distance in representation space.

The results show that dissimilarity-based sampling with $w_{tr} = 8$ and $k = 8$ achieves perfect discrimination. Reducing the number of selected windows to $k = 4$ while keeping $w_{tr} = 8$ preserves performance, whereas shortening the sub-trajectory length to $w_{tr} = 4$ degrades discrimination. Full-trajectory sampling is the weakest condition, supporting the conclusion that adaptation over all consecutive windows can overfit to global temporal cues. Random sampling improves over full-trajectory sampling but remains less reliable than dissimilarity-based selection. These results indicate that both semantic diversity and sufficient local temporal extent are important for discriminative test-time adaptation.

F.2 EFFECT OF TEMPORAL MEMORY ON VALUE FUNCTION ESTIMATION

Table 6: Validation VOC scores for progress estimation under distribution shifts. ID = In-Distribution, ES = Environment Shift, EM = Embodiment Shift, ES+EM = Environment and Embodiment Shift.

Shift	Dataset	TTT-TR	TTT-RS	TTT-EX	VITA
ID	tk_pnp	0.1917	0.1918	0.1957	0.7822
ES	lm_pnp	0.1843	0.1854	0.1826	0.7246
	td_fold	0.1402	0.1393	0.1322	0.7085
	ft_fold	0.1302	0.1311	0.1683	0.6583
	rd_fold	0.1161	0.1172	0.1154	0.6056
	ms_sweep	-0.0225	-0.0191	0.0102	0.4898
EM	dt_tk_pnp	0.2123	0.2118	0.2069	0.8203
	dt_tk_stack	0.0833	0.0830	0.0747	0.7081
ES+EM	dt_ft_stack	0.0558	0.0537	0.0439	0.6979
	dt_rd_pnp	0.1665	0.1660	0.1559	0.6951

We compare four adaptation strategies: (1) TTT-TR, which applies a single offline gradient update computed over the full trajectory; this batched update averages losses across frames and removes temporal order; (2) TTT-RS, which resets the adaptation module at every timestep and updates using only the current observation, yielding a memoryless estimator with no temporal carryover; (3) TTT-EX, which resets the adaptation module at every timestep but updates from a local window of recent frames matching the meta-training window size ($w_{\text{tr}} = 8$); the gradient average over this window does not preserve within-window order; (4) VITA (OURS), which updates the adaptation module incrementally at every timestep and therefore forms implicit memory through sequential parameter accumulation. We apply the same test-time hyperparameter sweep to all variants, as described in Appendix D. The optimal setting uses learning rate 0.1 and one adaptation step for VITA, TTT-RS, and TTT-TR, while TTT-EX performs best with learning rate 1.0 and one adaptation step.

Table 6 shows that TTT-TR, TTT-RS, and TTT-EX obtain similar VOC values. Thus, batched adaptation over either a short explicit window (TTT-EX) or the full trajectory (TTT-TR) behaves comparably to the memoryless baseline (TTT-RS). By contrast, VITA consistently outperforms all three alternatives, demonstrating that sequential per-timestep updates are necessary for effective value estimation in these real-world manipulation tasks. For the implicit-memory formulation, updating from a local window would double-count recent observations: they would enter explicitly through the window and implicitly through the accumulated parameters. Updating less frequently would also omit intermediate frames and lose temporal information. The implicit-memory TTT mechanism therefore requires sequential, per-timestep updates without reset.

G COMPLEXITY ANALYSIS OF DISSIMILARITY-BASED SAMPLING

Let $\mathcal{W}$ be the candidate window set with cardinality $N_c = |\mathcal{W}|$. Exact diverse-subset selection as in Eq. 4 requires enumerating $\binom{N_c}{b}$ subsets, so we use a scoring heuristic in which each candidate receives the sum of its pairwise distances to all other candidates, i.e., the row sum of the distance matrix. With $N_c = T - w_{\text{tr}} + 1$ candidate windows, each flattened window has dimension $m = w_{\text{tr}} \cdot d$, where d is the dimension of a multimodal representation z_t . We compute the full pairwise distance matrix $D \in \mathbb{R}^{N_c \times N_c}$ on the GPU using batched `cdist`, which has operation complexity $\mathcal{O}(N_c^2 \cdot w_{\text{tr}} \cdot d)$.

In our experiments, training trajectories are short, with mean length 69 and maximum length 120. For batch size 32, feature dimension $d = 1024$ for the CLIP-based multimodal representation, and window length $w_{\text{tr}} = 8$, the worst-case computational cost is

$$32 \cdot N_c^2 \cdot w_{\text{tr}} \cdot d = 32 \cdot 114^2 \cdot 8 \cdot 1024 \approx 340 \text{ MFLOPs.}$$

This cost is negligible relative to one model forward–backward pass and provides the diverse-window selection used during training.